

Technoscience and public understandings: the British case *c.*2000

IN THIS CHAPTER, I LOOK back on the histories I have presented, and I look around at the condition of STM. I present a view from Britain *c.*2000, partly because it is difficult to get a more general view, but also because of the interest of the exercise. Other countries will look different and that will influence the way that readers elsewhere use this book. Again I use my local examples partly to emphasise the fact that we all live (partly) in localities, and that general histories can help us 'place' those localities. I use the now common notion of 'public understanding of science' to ask questions about the present social meanings of STM, and I develop some suggestions as to how 'ways of knowing' might contribute – both to the analysis of problems and to the construction of plausible remedies.

In the twentieth century as presented here, STM became increasingly dominated by technoscience – by the industrial-academic-governmental networks that manufactured 'knowledge-based' commodities. As we noted, the technoscience of the post-Second World War decades was in large measure a creation of national governments, especially the military. Since the 1970s, governmental influence has decreased (not least in the former Soviet Union) and commercial interests have become more concentrated and more global. Though the nuclear project may have declined somewhat, the electrical-informatics complex has grown apace, and so have the major pharmaceutical companies. Both complexes are offering to change our lives. The decades around 2000 are being presented as *revolutionary* – like the Industrial Revolution around 1800. Science seems full of

transformative potential, and in that perspective *consumer resistance* now seems a major problem.

Here we approach what several historians have seen as the paradox of twentieth-century STM, especially medicine. Our understanding of the world and of our bodies is immeasurably deeper than in 1900; we have many techniques and remedies which were then only dreamed of, and many which were then inconceivable. And yet science and high-tech medicine are mistrusted, maybe more than in 1900, certainly more than in 1950 (Porter, 1997).

As we all know, this 'paradox' has many components. Western publics are much less obedient to authority than in 1950 – that is evident in matters of law, interpersonal behaviour, religion, etc., as well as for science. In 'matters of opinion', if not financially, British society is more egalitarian and less hierarchical. The media, whose 'discretion' could be relied upon in 1950 by governments and professional bodies, now place a premium on 'exposures' and investigative journalism; for better or worse, newspapers can sell copy by presenting genetically modified plants as a danger to health. From the 1960s, 'minority groups' that had been marginalised or subordinated have campaigned for their rights; the women's movement, gay rights, and campaigns for the new 'ethnic' communities have made Britain much more pluralist than it was. In these respects Britain reflects much of the West, and indeed has followed patterns first seen in the USA.

In social politics *and* in STM, the twentieth century was in large part *the American century*, and there especially social politics were linked with questions about the authority of STM. Women objected to being but 'mothers' in a medical system still oriented to the values of (male) production and (female) reproduction that were central to state medicine through the first half of the twentieth century. Homosexuals rebelled against their medicalisation, and blacks against the hierarchy of race which medicine had reflected and legitimated from the nineteenth century. For 1960s radicals, the 'consensus' politics of the 1950s and the high regard for professionals, especially scientists and doctors, suggested conformity – even social control.

Some writings on medical history came directly from this ideological realignment, notably the work of Foucault in France, and the associated studies prompted by the anti-psychiatry movement in the UK and the USA. Writings on 'ecology', notably Rachel Carson's (1962) *Silent Spring*, helped create a new movement for the defence of the environment (Bramwell, 1989; Sheal, 1976; Worster, 1994). Romantic attitudes to nature were resurgent and combined easily with a popular 'orientalism' in the promotion of alternative medical systems.

Generally, as in many such restructurings of popular opinion, new perspectives from the young and from 'outsiders' interacted with historical contingencies. Accidents in nuclear plants, which in the 1950s had seemed but technical hitches, now assumed a higher profile, partly because of campaigns against nuclear weapons. The thalidomide tragedy, when journalists discovered that babies had been born limbless as a result of medicines taken during pregnancy, shook public confidence and forced major changes in the regulation of drugs (and a major escalation in the complexity and costs of clinical trials). By the late 1960s, the two bright emblems of post-war STM – nuclear energy and wonder drugs – were casting long shadows. In the USA, the movement against the Vietnam war mobilised students and young academics, calling into question the military-industrial-academic complex which had been so important for the funding of STM.

This counter-cultural movement profoundly changed the social politics of the West, undermining the prestige of orthodox STM. But even in France, where student movements coincided with labour unrest and shook the state, the institutions of government (and of commerce) were not radically changed. The post-war consensus was, however, further weakened by the economic recession of the 1970s, especially the oil crisis of 1974 which revealed the vulnerability of Western economies to challenges from the Middle East and from the rapidly growing economies of the Far East. The political beneficiaries were the neo-liberal right, rather than the Romantic left.

In Britain, the radical critique of professional and governmental expertise was taken up by proponents of the free-market – an

ideology which had been worked underground for thirty years; Mrs Thatcher and Mr Reagan wanted to roll back the state, and they succeeded; they looked to 'business values' rather than the traditions of public service, to accountants rather than civil servants. Mrs Thatcher reasserted Victorian values – of independence, philanthropy and the family – and denied the existence of 'society'. The move to the right in Western politics from c.1980 led to 'privatisation' of many state enterprises, including research laboratories, and the 'franchising' of governmental functions to 'businesslike' organisations. It greatly encouraged the growth of 'business management' as an ethos and practice; American management consultants were evident around British government from the early 1970s. Their kind of formalised instrumentalism – the mission statements, targets, audits, etc. – had become central by the 1990s, pushed into what remained of the public sector, usually against the wishes of state-employed professionals such as doctors and academics.

I recall the long faces of colleagues accustomed to advise governments on 'science policy' when they returned from a meeting in London in the mid-1980s: government ministers had declared that Britain had an 'information mountain', like the notorious 'mountain' of butter then surplus to the needs of the European Economic Community. To the usual claim that research was essential for future industrial success, Thatcherites had replied that Britain's problem lay in knowledge application rather than knowledge generation. At that time, too, universities were an easy target for expenditure cuts, both because of the recent history of 'unrest' and because they faced a demographic decline in the numbers of eighteen-year-olds.

This double threat to publicly funded 'science' from both sceptical publics and neo-liberal governments provoked some scientists to mount campaigns to improve the 'public understanding of science'. Their collaboration with sympathetic journalists, politicians and scientifically educated industrialists is perhaps best understood as a 'social movement', chiefly concerned to secure state funding for scientific researchers and to protect that enterprise from adverse public criticism. But their

concern with the support of science has linked easily with the promotion of new technologies and with demands for improvements in scientific and technical education. It offers a useful point of entry for analysing present public meanings of STM and relating them to the concerns of the book.

'No one understands us'

Worries about the place of science in popular culture are hardly new. In the 1830s, young 'analysts' pointed to a British 'decline' as part of their campaign to gain the state support for science then evident in France and Germany. In the final third of the nineteenth century, a variety of enquiries, often prompted by proponents of science, argued for strengthening scientific and technical education at various social levels. Around 1900, 'science' was central to campaigns for national efficiency and the journal *Nature* was campaigning to raise the profile of science among the governing classes. After the First World War, while some social groups worried that science and technology were advancing out of control, others, especially on the left, came to see the rational application of science as the chief means to cure social ills and increase welfare at home and abroad. Debates continued after the Second World War – usually covering much the same ground as earlier debates (but without much sense of historical continuity). Problems were presented as newly discovered yet deep-seated; but one debate, from the 1950s and early 1960s still resonates (Alter, 1987; Edgerton, 1996b).

When the scientist-turned-novelist-turned politician, C. P. Snow, complained that the British ruling classes remained woefully ignorant of science, he was set upon by F. R. Leavis, the puritanical leader of a school of literary criticism. Leavis looked to literature, especially Victorian novels, as exploring and (thus exemplifying) the moral qualities which must remain central to public life. Like Matthew Arnold and many continental intellectuals, he saw sciences as 'instrumental,' as knowledge of *means* rather than *ends*, and thus as accessory rather than central to moral and political

culture. This was common nineteenth-century formulation, as we saw in Chapter 2, but in post-Second World War Britain the reaction against C. P. Snow would seem to have been energised by the obvious fact that the majority of government investment in post-war universities and colleges went for science and technology rather than the arts. That preference has continued as central to educational rhetoric, and there is often embarrassment over the way that student preferences (and job markets) have chiefly benefited undergraduate education in arts and social sciences.

But if Snow intended to extend the political power of the sciences into a cultural hegemony, then he failed. The debate prompted a few initiatives in liberal education; several universities developed history of science or science-studies programmes in the 1960s and early 1970s explicitly to equip science and technology students for leadership (or political) roles. These programmes continue to be influential (and I acknowledge my own debt to the investments made in Manchester), but their best intellectual products were more critical than technocratic, part of the public scepticism about scientific and medical authority which mounted from the late 1960s.

To measure the failure of the technocratic ambitions of the scientific and engineering communities, one has only to compare them with the astounding rise since the 1970s of management sciences and various kinds of accountancy. It is *this* kind of technical training, offering generic, 'transferable' skills which has succeeded, not just in the worlds of business, but in public service institutions such as government departments, cultural agencies and universities. This *managerialism* has indeed established itself as the general education of the late twentieth century. Students in many disciplines are encouraged to do management courses, where in the 1960s they were encouraged to include 'liberal' options. The rise of management seems to be worldwide, and contests of managers with liberal professionals (e.g. doctors) or creative professionals (e.g. television producers) are central to contemporary redistributions of power. It is against this background that we must view the changing fortunes of 'public understanding of science'.

Science back in business

The public-understanding movement has continued to the present, but much has changed since the 1980s. The continuing 'globalisation' of business, the rapid development of the techno-scientific complexes around informatics and pharmaceuticals, and the projection of our age as a new industrial revolution have 'clarified' the roles expected of 'academic' science. The promotion of business methods across the public sector, including universities, has created a culture of 'output', of knowledge as commodity, which spreads the values of industrial science across the whole academy, even to the humanities. Whereas in the early 1980s scientists and doctors had felt deserted by government and in need of friends, now they know the roles in which they can expect to succeed. They are not to see themselves as advancing disciplinary knowledge, nor as representatives of a public interest, nor as educators bringing the best out of students; they are creators and managers of knowledge commodities.

More and more, the British government is now managing universities to reduce input and increase output, and the procedures are crude. The research councils which award public funds for research projects often see themselves as part of the infrastructure of national production; even social research has been managed primarily in support of economic development. In the 1990s, universities had to press hard to include community benefits as an alternative goal for research (and, indeed, for a university funding system focused heavily on student numbers and research publications). During the 1990s, as the medical charities emerged as a major force in British research funding and policy, they too became more concerned with commercial production, especially around genetic engineering. Via intellectual property rights, the charities expect financial returns on research investments.

But if these complex, rapid shifts over the last twenty years of the twentieth century 'clarified', for better or worse, the roles of the British academic scientist, they did not solve the problem of 'public understanding'. Indeed, as science has become more

closely identified with business on a global scale, so sections of the public have become even more suspicious. Once they feared that science was endangering them because of some internal dynamic – some reckless pursuit of power over nature; today they fear that scientists are in cahoots with big businesses – messing up the world in pursuit of profit. So now we find worried business people, industrial scientists, leading academics, charity managers and government ministers of trade conferring and collaborating to ‘sell’ science. Public understanding of science has become a corporate good and a corporate goal.

The continuing campaigns for science and the associated hype around new industries, appear to be part of the reason for the growing currency of popular science – as books and on radio and television. Awards for scientists, journalists and playwrights, and meetings sponsored to increase interactions between scientists and writers have helped create a climate where some writers see science as part of their engagement with a dynamic present, and scientists see popular writing as a public service and a reputable career strand – no longer a vulgar distraction. Scientific and medical titbits are being introduced into news magazine programmes as matters of common interest, and the media can now rely on a dozen or so high-profile scientists with well-honed presentational skills. Scientists aspiring to that status or worried that they may be drawn defenceless into public controversies, can learn those skills in dedicated training programmes. In such a context, public interest in science, or public understanding of science, is not to be read in terms of general ‘levels’, as if it were the water in a bath fed by a ‘trickling down’ from scientists and by natural ‘upswellings’ of public concern. Better to see public concern as contested ground, where organised lobbies do battle, and in which journalists of various kinds may grind axes as they look for good ‘angles’.

Surely such debate is to be encouraged, and it should be enriched by commentators who can elucidate the issues involved and weigh the arguments on each side. That is one responsibility of academics concerned with science studies, whatever their personal attitudes and preferences. To that end, therefore, it is

important that funding for science studies not be dependent on the public relations lobbies for science, or indeed on anti-industrial lobbies. Part of the worry, of course, is the asymmetry here: environmental and public defence groups are paupers compared to the scientific industries and research charities. Moreover, as we have discussed, the latter interests are directly influential within government and in the state's higher education machinery, including those agencies which fund most of 'science studies'. We need to ensure that the public interest in disclosure and adequate analysis can continue to be satisfied by the scholarly community as well as by campaigning groups; that is part of the infrastructure required for a democratic politics in technical domains. But how shall we understand the ways in which 'publics' read 'meanings' in such fields?

The study of 'public understanding of science'

Academics who study in this area still operate by distinguishing an old view of the public from a 'new' one. The former was concerned with measuring how much 'science' was known by members of the public. Measurement was usually by interview or questionnaire; respondents were asked whether the sun went round the earth or vice versa. This view of the public is related to what the philosopher, Karl Popper, used to call 'the bucket theory of the mind' – minds were containers, so how much science, in the sense of 'scientific facts', did 'public' minds contain? By contrast, the (not so very) 'new' view regards minds as active rather than passive. All people, even laypeople, interrogate their environments on the basis of previous conclusions and assumptions – their hypotheses if you like. This approach to public understanding stresses the activity of the knowing person, and the way in which people become expert in that which directly concerns them. In this perspective, whether the earth goes round the sun or vice versa is not very important for members of the public – knowing the answer is little more than a trivial pursuit. But whether fumes from a chemical factory will

damage your garden, or whether food additives will damage your baby – these are questions that matter, and which laypeople will pursue with vigour and acumen.

Note that so far in this section, public understanding of ‘science’ means public understandings of the world as interpreted by scientists. There is, of course, another set of meanings carried in the ambiguous slogan – it could mean the public understanding of ‘science’ as an activity of scientists, or of ‘science’ as a community of investigators.¹ How do the old and the new views of public minds relate to that ambiguity?

Under the old view, laypeople learn scientific facts more or less adequately from school, newspapers or television. They may also learn some facts about the methods of science, or about various scientific communities – such knowledge is on a par with the knowledge of nature, in that all is supposed *factual* and independent of the observer. Both kinds of knowledge are guaranteed by belief in the veracity of the reporting agency, i.e. ‘science’, whether the report concerns the physical world or that part of the social world that reliably reports on the physical. By contrast, if we assume that the public are enquirers, and see themselves as enquirers with a particular point of view, so they are likely to see ‘scientists’ as also operating within particular perspectives of their own. They may well see scientific findings as ‘*objective*’ but as *less than the ‘whole story’*. For example, even when patients’ groups work closely with specialist doctors, they may still feel that the facts which patients need are best obtained from other patients, that doctors do not have the experience of dealing with a particular illness day in, day out – ‘from the inside’.

Where discussion is polarised and opposing interests appear to be involved, then the ‘scientific facts’ become part of the dispute. Even if honesty and openness are evident, it will be understood that the production and presentation of facts is properly the object of scrutiny. Our antagonistic process then serves, hopefully, to maintain standards of evidence by facilitating expert scrutiny of claims by both sides. But for such debate to be fully effective it is also necessary that both sides are able to *create* evidence – by literature searches, interviews, analysis (and reanaly-

sis) and by experiment. As Jerry Ravetz (1971) has stressed, the most worrying aspect of public scientific debate is not the suppression of evidence for a particular proposition, but the *absence* of such evidence when the research has never been carried out. In this respect democratic governments may be said to have a peculiar responsibility to fund research which may question powerful industrial and scientific interests, including the interests of government itself. This is the scientific correlate of the responsibility to facilitate debate by supporting the historical and social research community in ways that put its skills and knowledge at the service of otherwise disadvantaged groups.

In the remainder of this chapter I will try to show how the typology and schematic history presented in this book may be helpful to the various constituencies involved with such debates. Broadly speaking, I trace a path back over the concerns of the book – from technoscience back to meanings. First I discuss the roles of commerce and of public interest within technoscience, and especially in universities; then I use ‘analysis’ (and experiment) to reflect on the demarcation of ‘scientific’ expertise. I use ‘natural history’ to explore present and potential public roles of STM; and finally I return to questions of meaning and value.

The politics of technoscience

If the analysis of technoscience presented here is remotely accurate, then it would seem to follow that public understanding (and control?) of technoscience should be high on the political agendas of most nations. That conclusion seems to be borne out by the prominence in the media of many of the STM issues touched on here – the politics of health and health services, reproduction and ageing, food safety and environments, conservation and climate change, energy and armaments, etc. Adequate discussion of these issues and their connections would take a whole book, much longer than this one, but perhaps this is the place for some suggestions about modes of analysis.

If we think of technoscientific complexes as variously compounded of universities, companies and government agencies, then we can ask about the relative powers and prerogatives of the parties involved, and how they change. Since there seems little doubt that commercial interests are growing stronger rapidly – by direct power, through shifts of boundaries between private and public sectors, and by the private ‘colonisation’ of the public sector – the benefits and costs of that restructuring need to be examined. A sample of particular issues may illustrate the general problem.

One is the question of intellectual property in natural species, especially in the Third World. We have seen in Chapter 3 that exploration for commercial purposes was a major component of the growth of natural history and that imperial powers mounted expeditions and amassed collections for commercial reasons as well as for the demonstration of political and cultural power. The links between science and colonial exploitation are long established – but the recent developments pose new problems as well as old. In as much as species are being patented, they are appropriated as well as exploited; they are taken out of the public realm. Third World countries and their supporters argue for vesting the potential ‘property’ in the countries involved, at least for the purposes of preventing appropriation.

Similar issues have arisen over the patenting of sections of the human genome. Biological entities that most people presumed to be inalienably ‘public’ have been privatised. The extent of privatisation of the genome may have been diminished by the acceleration of public and charity investments in genomic analysis – thus limiting the amount which private companies are able to control – but one may wonder why the genome cannot be protected by adjustment or reinterpretation of law so as to exclude the patenting of natural entities. The boundaries of what can be patented have varied considerably over time and between countries; they could perhaps vary some more.

These issues have a history. In Britain after the Second World War there was much discomfort over the American patents on the preparation of penicillin – because it was a British discov-

ery, made by doctors and scientists in the public sector, and because it was a naturally occurring substance urgently needed by millions (Bud, 1993; 1998; Macfarlane, 1979; 1984). In our present, part of the offence arises from the knowledge that the vast majority of the work that made possible the human genome project was funded by taxpayers and conducted by scientists in the public realm, who are now to pay for the use of discoveries.

Through patenting and the many forms of sponsorship, the tradition of public service research is now seriously endangered by commercialisation of university science. Perhaps our need now is less for *public understanding of science* than for a much wider and stronger *understanding of science for the public* – the roles that STM could and should play in the defence and development of public interests. In both the above cases and in others we have mentioned, the problem is to find mechanisms for the defence of the public realm against undue intrusion from commercial interests.

One answer is to refine and extend the regulatory and informative roles of government. In sectors where life was at stake, such as the safety of medicines, most advanced countries now have reasonably effective mechanisms. In some newer areas, such as the regulation of reproductive techniques, Britain has pioneered agencies which appear to work well in as much as they command public consent while keeping the issues at a distance from main-line party politics. It seems likely that public opinion will force the creation of more and stronger protection agencies, though they are usually resisted by producer interests.²

Such agencies typically involve expert professionals and laypeople, and they are open to scrutiny by the public, the media and campaigning groups. In as much as the Internet etc. can serve to make them (and their results) more accessible, and possibly to reduce the cost of duplication of enquiries across nations, they would seem to be a very promising part of the future of STM. How such protection can be extended to poorer countries, perhaps by United Nations action or by levies on major companies, remains unclear, except that we might try to think of global

systems in which such protection would be as 'natural' as free trade is now claimed to be.

In as much as governments can be held to account for public protection against adverse effects of STM on individuals, we are likely to see extension of these regulatory roles, including agencies to ensure the probity of scientific publications now that huge sums may depend on getting the correct results or on disguising failures (Kevles, 1998). When more and more scientists are financially involved with commercial companies, the operation of peer review by research councils and charities may need public scrutiny, including public registers of interests such as are now obligatory for parliamentarians. One of the advantages of thinking across the range of STM may be to facilitate comparative study of the various forms of regulation – across sectors, functions and nations. Certainly, we need more – and more public – studies of this kind.

But governments are also involved in technoscientific complexes as purchasers. Here, too, Britain, following the Netherlands offers some encouraging examples. Agencies have now been created to assess new drugs and other medical technologies, not just for safety but for effectiveness and for cost benefit compared to others treatments. Presumably there is scope for much international flow of assessments of this kind, and much benefit to other potential purchasers, including individuals. Perhaps such agencies might be seen as continuous with the assessments (and consumer satisfaction data) collected by consumers' associations. Maybe the support of consumer information systems will come to be seen as a major governmental responsibility – part of the 'natural history library' required for effective living in a highly complex commercial society.

All such agencies allow better public and private choices; they would seem obvious candidates as potential commissioners of the public sector research which we argued above was essential for proper public defence. Charities such as Greenpeace or Friends of the Earth, or user groups in medicine, or local charities concerned with particular issues might also be appropriate. By widening the range of interests and groups that might command such research, we may balance the intense present

pressure which equates the public interest with economic development. Moreover, similar arguments could be made about Third World interests. Whereas once British anthropology was supported in the hope of improving colonial rule, now our 'overseas research' is strongly focused on potential markets. We can surely afford a wider definition, even of 'British interests'.³

But in this discussion of commercial and public interests, where do universities fit – for their teaching, and especially for their research?

Understanding public science

In previous chapters, universities have often been mentioned but their governance and their legal forms have not been discussed. We have presented them as most people now see them, as independent agencies receiving money from taxes but traditionally at arm's length from government. Of course, like so many traditional views, this one needs to be understood historically and in cross-national context; and we need to recognise that universities are now in flux. Historically, the German model is crucial for two features – the ideology of research as a human good, and the productive interaction between the pursuit of knowledge and the education of scholars.

In Germany, universities have always been state institutions; the freedom of university staff to teach as they wished and of students to move between universities were key features of the nineteenth-century reforms – part of the ideology of research to which I have several times referred (Proctor, 1991). The usual image of the German system since 1800 (excepting the wars and the years of National Socialism) is of self-governing institutions competing for staff eminent in research – a system of competition and emulation which encouraged and supported talent and led to world eminence in most fields of research. Historians have questioned this picture, but present studies of nineteenth-century appointments tend to show that commercial and political 'interference' acted more to reinforce than to damage excellence. For

example, Turner has shown, as any academic might have suspected, that left to themselves some universities and faculties tried to appoint from their own staff and protégés, rather than importing talent. In such cases, the Prussian Ministry of Education protected the system against inbreeding (Turner, 1971). Other historians have shown that the pursuit of science was not as disinterested as sometimes presumed; when states set up a chair in chemistry, say, it was often with an eye to the improvement of local industry or agriculture (or medicine). But again, since it was usually assumed that prestigious chemists would be most effective economically, there appears to have been little conflict between commercial motivation and the appointment of staff with high reputations.⁴

The American system is a mix of educational charities (e.g. the University of Chicago) and state institutions (e.g. the University of Minnesota). From the later nineteenth century, the German ideology of research was central to all the major institutions, but it was combined with an American insistence on the practical. The usual result was a broad education in principles, followed by science-based professional training. American universities developed rapidly in the early twentieth century when wars and economic depression limited European advance, and after the Second World War the elite American universities, and especially the technical universities, benefited hugely from government research investment. The system has been rich and diverse enough to preserve a variety of ideologies, from corporate ally to pure scholarship (Geiger, 1997), but the commercial challenge is becoming acute – highlighted when a whole department within the University of California signed a deal to become, in effect, the research arm of a bioscience company.

In the UK, the oldest universities were religious corporations; the nineteenth-century foundations were mostly secular charities – self-governing entities, funded by donations and student fees, and jealous of their independence (Sanderson, 1972). State support for research only became important in the twentieth century, especially after the Second World War. Through the 1950s and 1960s, universities expanded rapidly and research

was well funded, largely on its appeal to fellow academics. Undoubtedly this meant that universities became less responsive to their localities – in terms of industry, public services and culture. The effect was accentuated by the ‘academicisation’ of the major technical colleges which had been funded through local government, but which were transferred to the university sector, culminating in the 1980s when all the former ‘polytechnics’ were brought under the same funding council as the old universities. The consequent wide disparities within the sector is one of the reasons for the elaboration of formula-funding mechanisms, for the diminution of support per capita (levelling down), and for systems of micro-review and heavy bureaucratisation.

I have suggested that increase in direct management and the stress on commercial values can both be regarded as aspects of the incorporation of the universities into networks for the production of knowledge commodities. What then is at stake, in terms of the maintenance of public service and public knowledge, and how might these public interests be defended? We have noted above that part of the solution could be a more open and diverse system of research support, where public-defence agencies and charities, etc. might commission short-term research, and might join with academics to put forward proposals for research aimed at public goods other than increased production and consumption. The membership and range of the research councils and research charities might need extending.

Widening the range of potential research funders, and reinforcing inputs from wider publics, might also help meet the need for expertise and experts who are not compromised by commercial involvements. But perhaps we also need to rediscover the value of learning and of research as an end rather than a means to more production or, even, safer environments. When the then ‘new universities’ of industrial Britain were founded in the later nineteenth century, they were meant to contribute to industry, but they were also meant to offer and develop values other than utilitarianism. The motivation derived, in part, from the German tradition of research; it overlapped with the drive for civic improvement or art galleries, as well as for better technical

colleges. In their STM, as well as in the arts and humanities, universities need to find ways of reclaiming that cultural heritage and those educational values. That may mean trusting professionals; it surely means encouraging the huge potential of universities for open-ended interdisciplinary dialogue, formal and informal. Universities as communities are being undermined in Britain because internal accounting and relentless 'reporting' occupy the space once allowed for creative conversation. Business management produces businesses.

A third general remedy is more experiment. To widen research demands, to recognise learning as an end, and to better serve local and global communities, universities and other public bodies need more room to be original. In the centres of most of the major cities of Britain, there are thousands of creative staff and tens of thousands of bright young people in institutions that are still largely funded by the public. If we want to be 'scientific' about developing regions and universities, let that 'science' include lots more experiment; 'scientific management' here should encourage institutional novelties, perhaps with rewards 'to follow'. Standardised systems of surveillance have the opposite effect.

In sum, it seems to me that the political problem now is not the *public understanding* of science, but rather the maintenance of *public science* – creating and using knowledge to best serve public interests. But we need to underline that this is an issue for *all kinds* of knowledge, not just for 'science' or STM. The limits of 'science', especially in the English language, are unhelpful. In this book, analysis and natural history clearly reach beyond the normal British boundaries of STM; perhaps we need to rethink those boundaries. We could probably benefit from thinking of STM more as we think of the social sciences and humanities (and vice versa).

Analysis and the bounds of 'science'

English-speaking countries use rather narrow definitions of science compared to countries that speak French or German.

'Science' in English is restricted in its subject matter – used alone it usually refers only to natural sciences, rather than to moral or social sciences; history and literary criticism are not normally called sciences at all. This 'width' restriction then has consequences for questions of 'depth' – or ways of knowing. It focuses our attention on the methods supposed to be most characteristic of natural sciences, i.e. laboratory manipulations, especially experimentation. Analysis of economic data, say, which would be 'scientific' in other languages, has an uncertain status in English, and the natural history of riverside meadows, at the level of recording species throughout the year, is unlikely to qualify as 'scientific'.

Both restrictions (and their interactions) seem unfortunate. As regards width, it seems discriminatory and divisive to routinely distinguish between two classes of systematic investigation on grounds of their subject matter. Critical analysis of economic and social phenomena is *prima facie* as useful and demanding as analysis of geological strata or ballistics. We need better analytical indicators of human well-being as much as we need better indicators of environmental change; we need informed debates about the limits of economics, as we need debates about analytical medicine. It is no accident that such forms of knowledge developed together, as we saw in Chapter 5. Recognition of that (historical) commonality could do much to overcome deep educational divisions that limit critical analysis of social phenomena, as they also limit critical attention to the operations and results of natural-scientific communities.

One reason for the British 'exclusion' of social sciences may be the undue emphasis placed on experiment rather than analysis as crucial to 'scientific method' – especially as popularised after the Second World War. By stressing analysis and the pluralism of science perhaps this book will make for more *inclusive* usages and attitudes. For example, it is customary for government ministers in debates about food safety to demand 'good science' as the basis for their decisions (or delay). Such calls are markedly less common in matters of social or educational policy, where the individual opinions of ministers or functionaries often appear as

sufficient authority for action. More symmetry might help – issues of safety cannot be decided only by ‘science’, at limit because they involve judgements about the ‘worth of a life’; social policy might benefit from rather more *analysis* – and *experiment* testing.

Indeed, social experimentation might be one of the beneficiaries of the ‘informatics’ revolution. I wondered at the start of this book whether new ‘ways of knowing’ might be added to my list, not just by creative critics, but through the evolution of STM. One possible ‘new way’ is *simulation*, which is already widespread in STM teaching ‘laboratories’. Some of the computing power once used for US military calculations is now available for civilian use, and capable of modelling the traffic patterns of a major city in full detail.⁵ There surely are possibilities here, especially for experiments in the redesign of institutions which anyway use programmes for many of their bureaucratic functions. Could we simulate a variety of systems for ‘experimental universities’ that would enhance public accountability and professional creativity, without a myriad of particular reporting mechanisms that partly overlap but are never substitutable?

Here and elsewhere, talk of ‘the scientific method’ tends to mislead. As Paul Feyerabend demonstrated, it is very difficult, perhaps impossible, to codify the rules which are followed in really creative science – that level of creativity usually means altering the rules (Feyerabend, 1975)! The real commitment is not to rules of method but to the exercise of *imagination and critical faculties* – and that applies as much in dramatic productions or the development of social services as in neurosciences – or it should. This book is built on the benefits of recognising the pluralism of STM and the continuities between different ‘subjects’ that can be recognised as ‘ways of knowing’. This is *inter alia* a tool for breaking down the supposed boundaries of ‘science’ and the supposed exclusivity of ‘scientific method’.

To that end, I now go on to suggest that including ‘natural history’ as ‘science’ may also be beneficial – not just for certain very necessary types of investigators (e.g. taxonomists) but for

the health of scientific enterprises and of publics generally. To include natural history is to include a wider public *within* the debate – as actual and potential participants, rather than as receptacles for expert assessments. The more we can see the *variety* of STM and its articulation with everyday life, the less likely we are to be mystified by technicalities. And at the end of the chapter, we move on from natural history to ‘meaning’.

Publics and natural histories

As we have suggested, natural history is often left out of ‘science’ and relegated as mere ‘information’. I have argued for a wide view of natural history, and for placing this descriptive, classificatory way of knowing at the heart of the scientific enterprise – as a cultural achievement on which analytical and experimental modes are built, and one which remains a major way of dealing with our world. In as much as natural history can be more easily learned and practised than analysis or experimentation, and in as much as it is more continuous with ordinary language and experience, it is more accessible to untrained publics. Natural history is often recreational and sociable; it is ‘phenomenally’ rich. Questions of amenity and/or safety, e.g. around pollution, are often to be researched at the level of natural history, and citizens have a right to ask for questions to be addressed at that level. Here I want to point up several ways in which our focusing on natural history can guide our thinking about public understandings of science.

Access and recruitment to science

There appears to be little social research on why students choose science (or why others avoid it).⁶ Much would seem to depend on public perceptions of ‘rewarding careers’; these, of course, are part of social history, and could usefully be so explored. Let me illustrate by personal anecdotes. In the late 1950s many of us chose ‘science’ as the default option at school – if you were bright and

could do science, you did so. Only if you were innumerate, or uncommonly passionate about history, say, did you choose to spend your final school years (and therefore your university career) doing languages or humanities. We knew about the Russian satellite 'Sputnik' that had so jolted the West, we had been visited by university professors who lectured us on Britain's need for scientists, and many of our parents had heard of someone who had 'done well' as an industrial chemist, or some such.

Of course, there were also more positive and subject-specific reasons. Some chose science because they wanted to do medicine. Others were fond of some aspect of science that they followed or practised 'out of school'. Maybe you were interested in debates about evolution, or went bird-watching, or you had a microscope, a telescope or a chemistry set, or you made radios or took photographs and developed them yourself. These were the 'hobby' roots of many scientific careers. For the most part, they qualify as natural history – collecting, describing, arranging, or crafting apparatus to produce 'special effects'. Sometimes these activities were solitary pursuits, sometimes a bond between friends, sometimes pursued with local societies – e.g. for astronomy or ornithology. Such interests undoubtedly shaped the choices and the inclinations of scientists, as Jeff Hughes showed for British physicists (Hughes, 1998).

I studied biology partly from an interest in botany encouraged by a chemistry master (and amateur botanist) and by my maternal grandfather. At the start of the twentieth century, my grandfather had attended classes and field trips at the Burnley Mechanics' Institute, taught by an autodidact 'biologist', Ernest Evans, who wrote books on local history and natural history and was also known as a scientific poultry-breeder (egg-science again!). My Grandfather's brother went on from 'the Mechanics' to the Imperial College at South Kensington, and spent his working life as an entomologist in the colonial civil service. They sprang from a tradition of Lancashire textile workers escaping from the cotton towns to the surrounding hills (and maybe reading Wordsworth and Ruskin). Perhaps we need to see science as part of family histories as well as daily life.

Nowadays, 'popular science' is largely about/via computers. Young prodigies can make their fortunes; adolescents come to university with a facility in informatics that embarrasses ageing teachers. On the biology side, students come with developed interests in ecology or in the wonders of genetic engineering. The former are more likely to have direct, out-of-school experience – and they are more likely to be disappointed by the strongly 'molecular' orientation of most university biology departments. If science and technology departments want to increase the quality of input, or want students with interests other than technical-business, they could do worse than attend to public participation in sciences at the level of craft and natural history. Archaeology offers a model here: in Britain it is not a school subject, but it recruits well at university level partly on the basis of local and regional 'digs' which school students can afford (and via the corresponding coverage on television, radio and in magazines).

In recent years, most public discussions of science education have concerned assessment procedures and entrepreneurship. There is room for much more exploration of the roles of science education for individual lives and collective action.

Information and utility

In Chapter 3, I stressed the similarities between the natural-historical orderings of life's riches and our present concerns with superabundant 'information'. The technology now exists that can bring into any home or office, stocks of information beyond what could be found in any of the major libraries. Natural history collections, museums more generally and art galleries are all accessible virtually; so are data on diseases, medicines and environments. The Internet allows endless unpredictable access to the huge quantity of data specimens that individuals and groups throughout the West have seen fit to display; it brings a new world to armchair explorers, a new richness to our virtual 'environments'.

Natural history, however, is not just a matter of information or a logical or pedagogical prelude to other sciences – it is part